

Pyroconvection 3-day forecast – Tuscany & Catalonia – run 2026-07-21 00Z

Pyroconvection type – 3-day forecast (valid 2026-07-21 .. 2026-07-23)

Run 2026-07-21 00Z. Daytime hours 09/12/15 Z per day. Classifier: `pyflam.pyroconvection_type` (bulk-Richardson ABL, Bolton LCL, mixed-layer $d\theta/dz$, cap γ - θ , ABL-top RH; no surface CAPE), plus the new dry-pyrocloud `fireABL/decoupling` diagnostic.

Tuscany – FUEL-GATED (hybrid ICON-EU + ICON-2I 2.2 km gate)

Expected: classed only where fire power ≥ 10 MW/m on the real .lcp fuels.

Tuscany – POTENTIAL (atmosphere only, upper bound)

Catalonia – POTENTIAL (ICON-EU 6.5 km only)

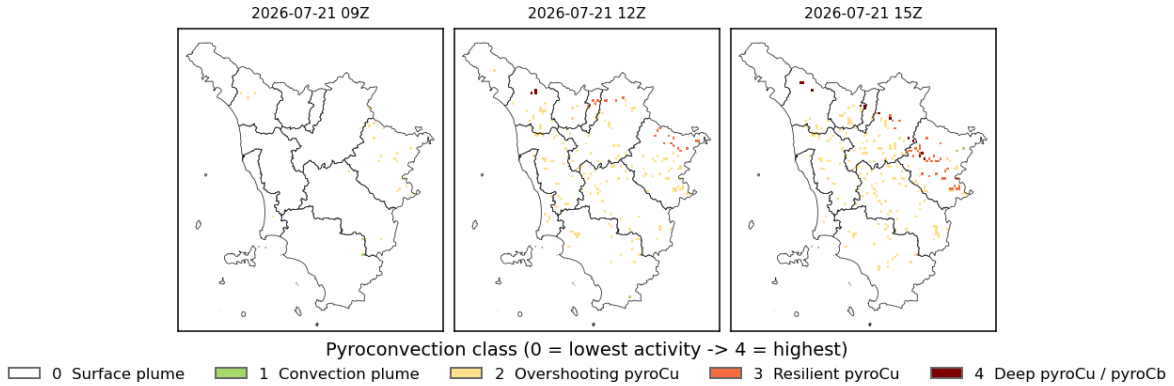
Catalonia is outside the ICON-2I 2.2 km domain and has no Catalan .lcp fuel, so only the ICON-EU-native potential (upper-bound) map is produced – no fuel gate.

Class scale

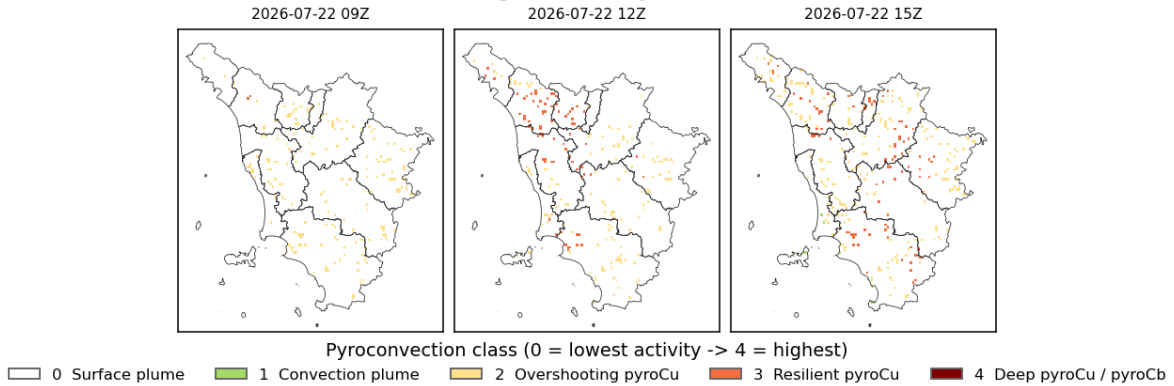
0 white	1 green	2 yellow	3 orange	4 dark red
Surface plume	Convection plume	Overshooting pyroCu	Resilient pyroCu	Deep pyroCu / pyroCb

Generated by `pyflam` (feat/dry-pyroconvection-fireabl). Tuscany hybrid via `tests/pyroconv_daily.py`; Catalonia via `scripts/catalunya_iconeu_3day.py`.

Pyroconvection type -- FUEL-GATED -- expected (only where fire power ≥ 10 MW/m on the Tuscany .lcp fuels)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-21 (run 2026-07-21 00Z)



Pyroconvection type -- FUEL-GATED -- expected (only where fire power ≥ 10 MW/m on the Tuscany .lcp fuels)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-22 (run 2026-07-21 00Z)



Pyroconvection type -- FUEL-GATED -- expected (only where fire power ≥ 10 MW/m on the Tuscany .lcp fuels)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-23 (run 2026-07-21 00Z)

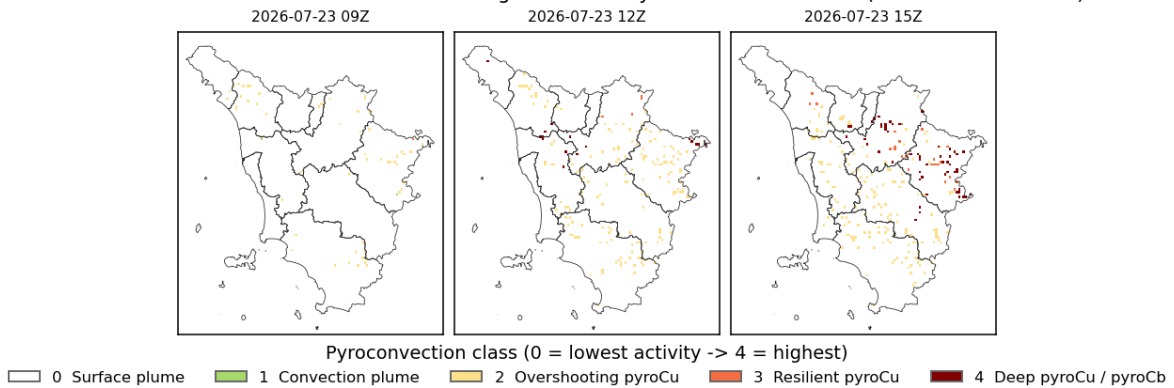
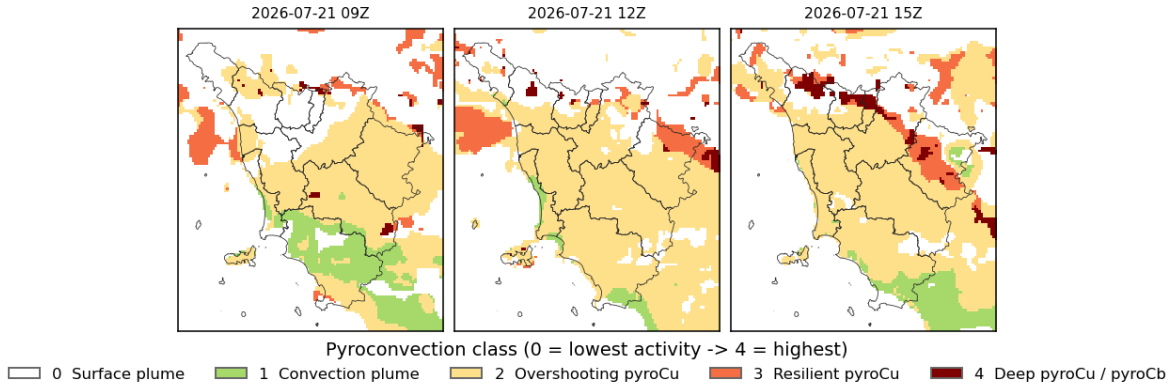
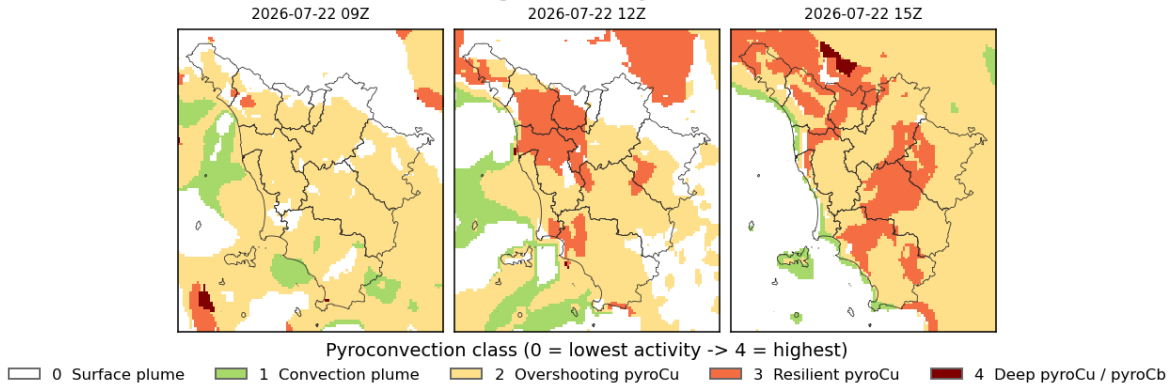


Figure 1: tuscanY gated

Pyroconvection type -- POTENTIAL -- atmosphere only (upper bound: assumes a pyroCu-capable fire in every cell)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-21 (run 2026-07-21 00Z)



Pyroconvection type -- POTENTIAL -- atmosphere only (upper bound: assumes a pyroCu-capable fire in every cell)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-22 (run 2026-07-21 00Z)



Pyroconvection type -- POTENTIAL -- atmosphere only (upper bound: assumes a pyroCu-capable fire in every cell)
 ICON-EU model levels + ICON-2I 2.2 km gate -- Tuscany -- VALID 2026-07-23 (run 2026-07-21 00Z)

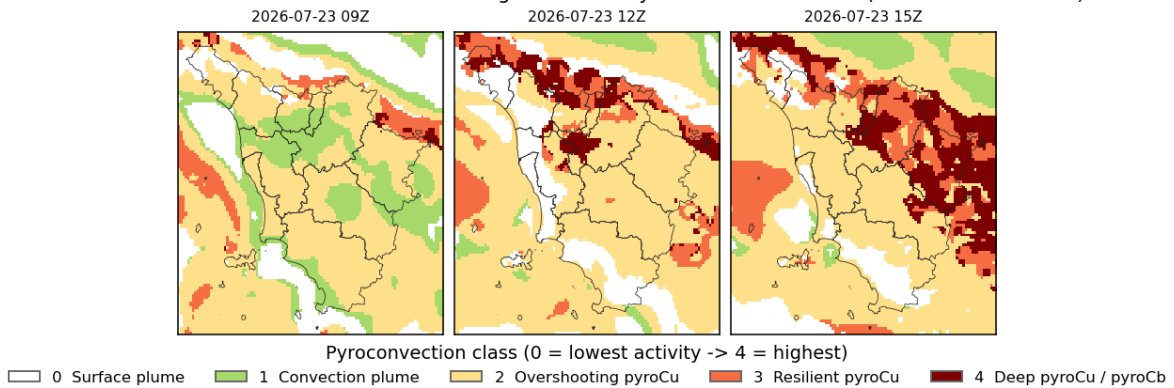


Figure 2: tuscany potential

Catalonia pyroconvection type -- POTENTIAL (atmosphere only, upper bound)
 ICON-EU 6.5 km model levels (DWD open data) -- run 2026-07-21 00Z

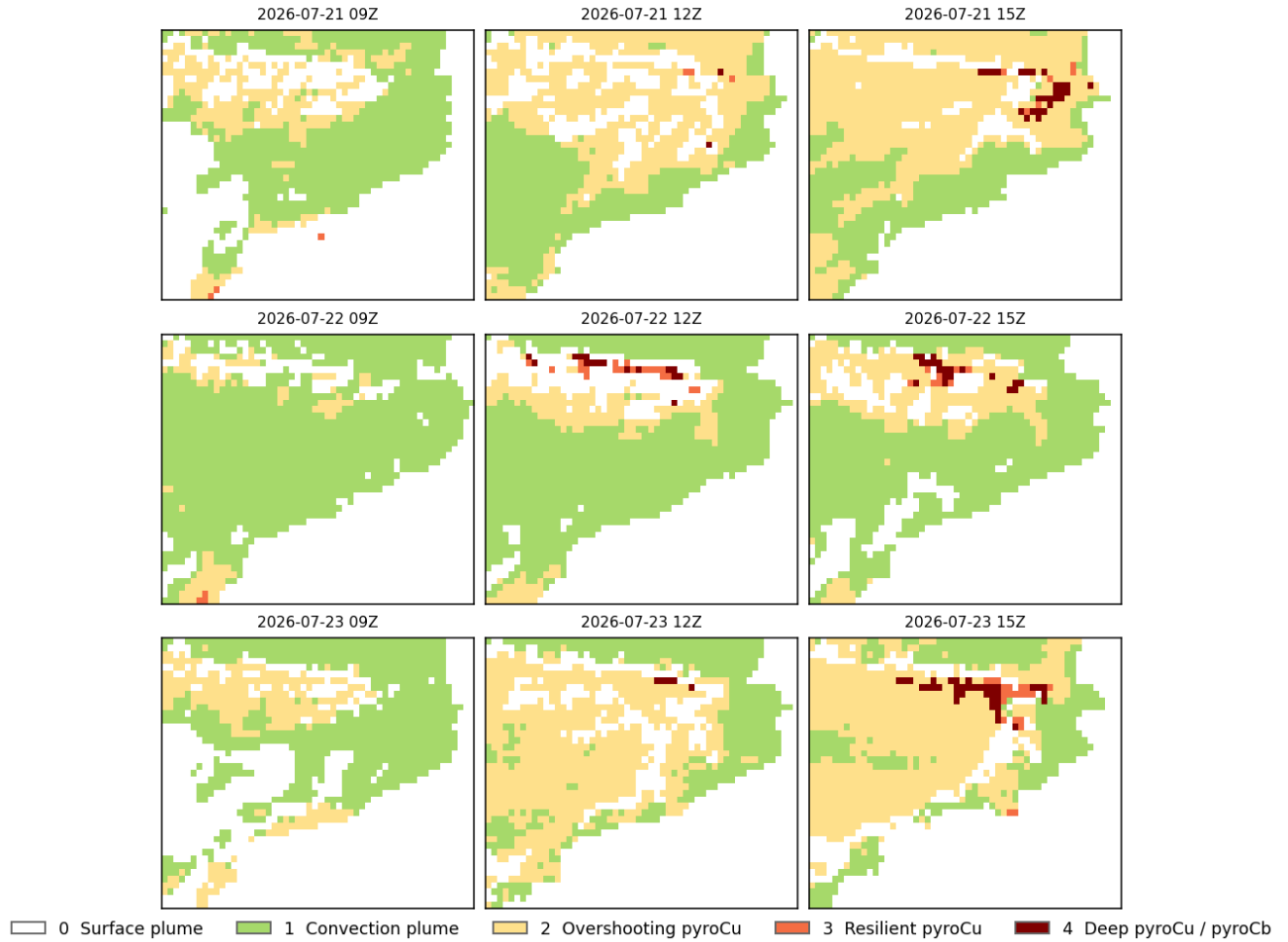


Figure 3: catalunya potential